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Roll No

CS-701 (GS)
B.E. VII Semester Examination, June 2020
Grading System (GS)
Compiler Design
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Construct a DAG for the basic block whose code is given below:
D: = B * C
E: = A + B
B: = B * C
A = E - D
2. Prove that Grammar is CLR but not LALR
 $S \rightarrow Aa / bAc / Bc / bBa$
 $A \rightarrow d$
 $B \rightarrow d$
3. Distinguish between top-down parsing and bottom-up parsing? What is the largest class of grammars that can be parsed by each of them?
4. What are the typical entries in a symbol table, what are the various data structures used to implement the table.
5. How does an Operator Precedence parser work? Use a pre-constructed operator precedence table to guide the parsing of an input 'a+b-20' using operator precedence parser.
6. Define a Quadruple. How is it different from triples? Convert the following expression into three address code and quadruple.
 $S = (a + b) / (c - d) * (e + f)$
7. Construct the DAG for the following basic block:
a : = b + c
b : = b - d
c : = c + d
e : = b + c

OR

What do you mean by LEX? Explain in detail.

8. Write short notes:(Any two)
 - a) Local and Loop optimization
 - b) Peephole optimization
 - c) Dead code elimination
